



GAME-BASED LEARNING IN ASTROPHYSICS: EVALUATING ORBITAL MECHANICS COMPREHENSION THROUGH SIMULATION

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Abstract:

Orbital mechanics is a fundamental concept in astrophysics, yet it remains a difficult concept for beginners to grasp due to its abstract and mathematical nature. Traditional teaching methods focused on theory often fail to provide insight into how objects actually move in space, limiting engagement and comprehension.

This paper investigates the effectiveness of game-based learning in improving understanding of orbital mechanics. A study involving 50 participants with limited prior knowledge of astrophysics was conducted using the astrophysics simulation game *Kerbal Space Program*. Participants were tasked with playing the game for two hours and subsequently rated their perceived improvement in understanding on a scale from 1 to 10. The average rating was 8.4, indicating a significant increase in comprehension.

Additionally, a custom-built prototype simulation was developed to demonstrate core orbital mechanics concepts such as gravity, thrust, and orbital parameters. The result suggests that interactive simulations are an effective educational tool for improving comprehension of abstract fields such as orbital mechanics and astrophysics.

Keywords:

Gamification, Orbital Mechanics, Astrophysics Education, Simulation, Game-Based Learning.

INTRODUCTION

Orbital mechanics is essential to understanding motion in space and is widely used in astrophysics and aerospace engineering. However, these concepts are often difficult for beginners to grasp due to the abstract and mathematical nature of their models. [1] It is common for learners to struggle to develop an intuitive grasp of how gravity and velocity interact to produce stable orbits.

It is essential for people to understand the core concepts of astrophysics in the modern age, as humanity relies on those scientific fields to keep our modern society functioning. Understanding the core concepts behind orbital mechanics and astrophysics helps further one's understanding of things such as telecommunications, navigation systems, scientific missions across the solar system, and other important pieces of space infrastructure we use every day.

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Game-based learning has emerged as a promising method for improving comprehension in complex scientific domains. By allowing users to interact directly with simulated systems, games like these provide an environment where learning occurs through experimentation and observation rather than passive instruction [2], [3]. Previous studies have shown that interactive simulations can significantly enhance engagement and retention in STEM education. [4], [5]

One of the most notable examples of this approach regarding orbital mechanics and astrophysics is the game *Kerbal Space Program*, an astrophysics simulation game that models real-world orbital mechanics. [6] See Figure 1.

Players are tasked with designing a spacecraft, launching it into space, managing trajectories, achieving orbit, and conducting advanced manoeuvres in space. As they completed their tasks, the players set various tasks of their own during their time playing the game. Nearly all participants managed to achieve a stable orbit around the planet, while some even landed on other celestial objects such as the *Mun*, which is the *Kerbal Space Program*'s equivalent of the moon of our Earth. This paper evaluates the educational effectiveness of such simulations and presents a small prototype designed to demonstrate the same principles in a simplified environment.

2. METHODOLOGY

To evaluate the impact of game-based learning on the specific field of orbital mechanics, a study was conducted involving 50 participants. All participants had limited prior knowledge of astrophysics and orbital mechanics. 50 participants were chosen as a study group due to that number being significant enough to provide useful data, while still being small enough for the study to be conducted. The participants were all students aged 18 to 26. They were tested remotely.

Each participant was given two hours to interact with the simulation game *Kerbal Space Program*. During this time, they were encouraged to experiment with launching spacecraft, creating their own, adjusting trajectories, and attempting to achieve orbit. No formal instructions outside the game have been provided to the participants, allowing them to learn through exploration and trial-and-error, which is a key principle in game-based environments. [7]

After completing the session, participants were asked to rate their perceived improvement in understanding orbital mechanics on a scale from 1 to 10, where:

- 1 indicated no improvement
- 10 indicated a significant improvement

The responses were collected and analysed to determine the overall effectiveness of the learning experience. The subjective ranking of 1 to 10 was used as a way to allow the participants to give their own subjective view of their own understanding, as, for the purposes of raising awareness of the core concepts of orbital mechanics and astrophysics, it mattered more to grasp the abstract concept of it, rather than have them take an objective test. Due to the factors such as the limited two-hour session per participant, focusing on additional advanced concepts was outside the scope of the study.



Figure 1. Launch of a Rocket in *Kerbal Space Program*



3. RESULTS

The results of the study showed a strong positive trend in perceived understanding of the subject. Participants had limited knowledge of astrodynamics, orbital mechanics, and astrophysics before the task. Upon finishing their session, the average rating among participants was 8.4 out of 10, indicating that most users experienced a substantial improvement in their comprehension of orbital mechanics.

Participants reported gaining a clearer understanding of:

- The relationship between velocity and orbit formation
- The role of gravity in maintaining orbital motion
- Understanding of real-life space missions
- The effects of prograde and retrograde burns
- The role of Δv in space manoeuvres

These findings are consistent with previous research which suggests that interactive simulations provide an effective means of teaching complex physical systems by allowing users to observe cause-and-effect relationships directly. [5], [8] The distribution of participant results is illustrated in Figure 2.

4. DISCUSSION

The results support the hypothesis that gamification can significantly improve a person's understanding of abstract scientific concepts. Unlike traditional education methods, which often rely on static diagrams and theory, simulation-based learning allows users to interact with dynamic systems which respond to their input. [3], [4]

Through experimentation, participants were able to develop an intuitive understanding of orbital mechanics. For example, players quickly learned that vertical ascent alone is insufficient for achieving orbit and that horizontal velocity was required as well. This type of understanding is difficult to achieve through theoretical instruction alone, especially in complex domains such as astrodynamics. [1], [9]

Additionally, the engaging nature of games encourages longer interaction time, which may contribute to improved learning outcomes. The combination of visual feedback, player control, and real-time system response creates a highly effective learning environment. [2], [5], [8]

However, it is important to note that this study relies on self-reported data. While the results are promising, future research should include objective testing methods, such as pre- and post-assessments, to measure the method more accurately.

5. LIMITATIONS

This study has some limitations that are worth noting. Particularly, the lack of a controlled experimental comparison between game-based learning and traditional learning methods. While participants reported a high level of perceived improvement in the topic, the absence of a control group means it is not possible to conclude with certainty if the observed learning gains are solely a result of the simulation-based approach rather than the general exposure to the topic. In addition, the relatively short duration of the experiment, which was limited to only a two-hour-long session, may not reflect the long-term knowledge retention with accuracy.

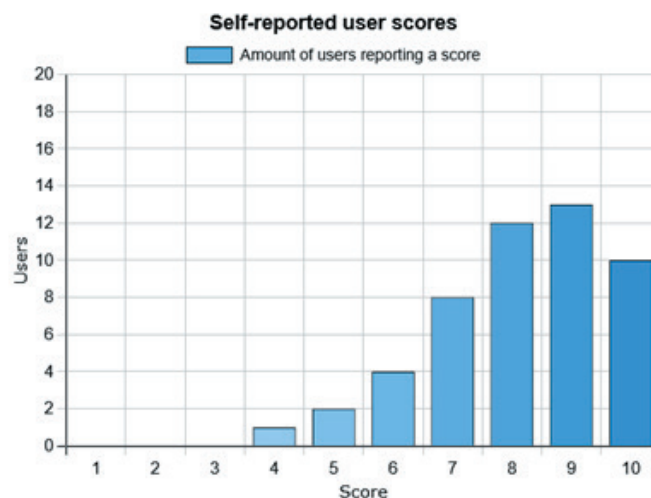


Figure 2. Self-reported user scores



It is possible that the participants developed an intuitive grasp of orbital mechanics concepts during gameplay; however, this understanding may diminish over time without further reinforcement.

Furthermore, individual differences between participants such as prior gaming experience, spatial reasoning capacity, and personal interest in related topics were not accounted for, which may have influenced the study's results. Participants who are more familiar with video games or have a natural inclination towards physics-based reasoning may have found it easier to engage with the simulation and extract more insightful information from it.

Finally, while *Kerbal Space Program* provides a relatively accurate, albeit down-scaled version of orbital mechanics, it still simplifies many real-world complexities, which may result in minor misconceptions if not supplemented with further education on the topic. These factors highlight the need for more comprehensive and controlled studies which are needed to fully validate the effectiveness of game-based learning in astrophysics topics.

6. PROTOTYPE IMPLEMENTATION

In addition to evaluating participants in *Kerbal Space Program*, a simplified orbital mechanics prototype was developed. The purpose of this prototype was to demonstrate that even a cheap, minimal system can effectively communicate core astrophysical concepts while keeping the cost of the project minimal.

The simulation allows the player to launch a spacecraft from a planetary surface and attempt to achieve orbit.

The system models gravitational attraction and allows the player to apply thrust in different directions to control the trajectory of the spacecraft. These mechanics follow the same fundamental principles described in classical orbital mechanics literature. [1], [9]

Key features of the prototype include:

- Simulation of planetary gravity
- Player-controlled thrust and orientation
- Orbital parameters such as apoapsis and periapsis
- Stage separation of rockets
- Time-warp functionality for accelerated simulation

To improve numerical stability, the simulation uses a spacecraft-centered reference frame. In this approach, the spacecraft remains fixed while the environment moves relative to it. This prevents floating-point errors that occur in large-scale simulations while maintaining accurate physics calculations. In addition, the simulation runs on a 1:100 scale of the solar system, with only major gravitational bodies included. This allows for a dynamic system where the user can experiment with properties of astrodynamics such as the planetary gravitational sphere of influence of different objects in space. The prototype has functions for generating a line showing the current trajectory of the spacecraft, this enables the user to see where their rocket is heading given the current motion and interactions with gravity wells. The time-warp function exists so that the player can fast forward the long waits between manoeuvres in space, due to limited control and the inaccuracy of controlling the spacecraft during time-warp, controls are disabled at speeds greater than 1x.

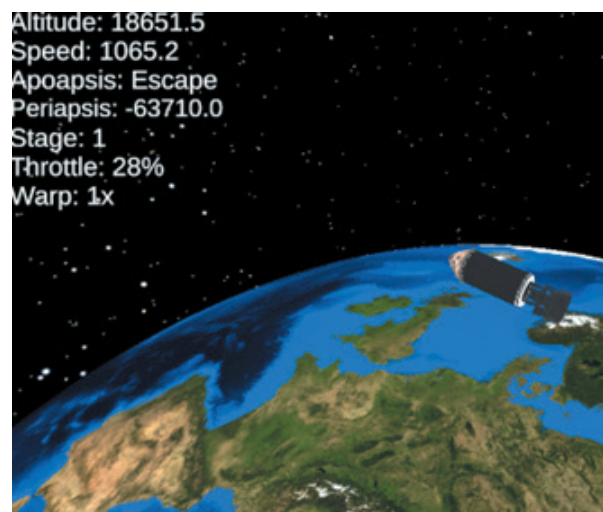


Figure 3. Custom orbital mechanics prototype showing a rocket in space above Earth



Stage separation of the rockets is modelled in a way that is similar to *Kerbal Space Program*, with the user able to detach stages upon pressing a key. If the stages themselves did not run out of fuel prior to the decoupling command, they will continue to burn in the direction they were pointing towards before decoupling. The they follow their own trajectory which adjusts according to their thrust, the gravity of objects, and the original trajectory. Due to floating-point errors, and the focus on the player-controlled spacecraft, detached stages get less precise as they distance themselves from the player; however, it is worth noting that this issue is fixable with additional development time.

Although simplified, the prototype successfully demonstrates how interactive systems can communicate the principles of orbital motion in an intuitive way. A visual representation of the prototype is presented in Figure 3.

7. CONCLUSION

This paper demonstrates that gamification is an effective approach for teaching orbital mechanics and astrophysics. The study conducted using *Kerbal Space Program* showed a high average improvement score of 8.4, indicating that participants gained a strong intuitive understanding of the basics of the subject.

The results highlight the importance of interactive learning in complex scientific domains. By allowing users to experiment with real-time simulations, games can bridge the gap between theoretical knowledge and practical understanding. [2], [4]

The prototype developed in this study further supports this idea by showing that even a low-budget simplified simulation can effectively communicate core concepts. Future work may focus on expanding this prototype and incorporating more advanced features, as well as conducting more rigorous evaluations using objective assessment methods in order to obtain a more objective result.

In addition, the findings of this study highlight the broader potential of integrating interactive simulations into formal education systems. As technology has become more accessible, tools such as game-based simulations could be incorporated into classrooms as supplementary learning resources along with traditional teaching methods in order to achieve better engagement with the source material and better teach abstract concepts to students. This approach is particularly effective in more abstract fields or those which are not instantly

intuitive to humans; in such cases, visualization and interaction play a key role in understanding concepts. The prototype built for this proves that the cost of such simulators does not have to be high, nor do they have to be very advanced to serve their purpose. Game-based simulators for such topics are required to be engaging with the user, and accurate so that the user can apply their theoretical knowledge in an interactive environment which reinforces the original source material. Furthermore, the increasing popularity of digital learning platforms suggests that such methods could be scaled and adapted for a wider audience, turning complex scientific topics into engaging simulations which can be more approachable to beginners.

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